Build a complete prediction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use one diagram or set of equations to connect net force, change in velocity, and final position.

SUCCESS CRITERIA - I can...

1. Connect force, acceleration and position consistently.
2. Show quantities and units at every stage.
3. Explain what the prediction treats as true and when it would stop working.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A **3 kg** box has μ_s=**0.50** and μ_k=**0.40** on a level floor; g=**9.8**.
1. **Compare pushes of 6 N** and **18 N**, starting from rest.
2. **Find actual friction** and acceleration.

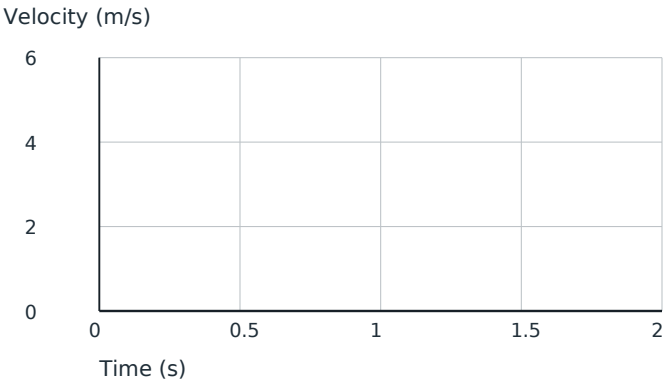
Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $a = \Sigma F/m$; $v_f = v_i + a\,\Delta t$; $x_f = x_i + \frac{1}{2}(v_i + v_f)\Delta t$. Constant acceleration; use one frame. i = initial; f = final. Apply each direction separately.

Worked reasoning: A **2 kg** cart starts from rest under **6 N** right and **2 N** left for **2 s**. Net force **4 N** gives a=**2 m/s²**. Its displacement is ½×**2**×**2²**=**4 m** right.

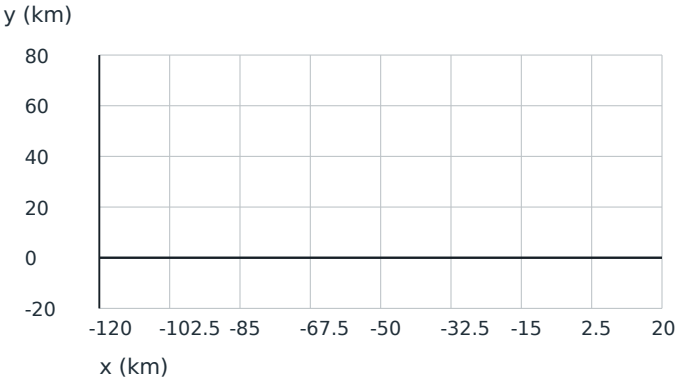
Your turn: 1. **Draw the force diagram** and the matching velocity-time triangle.
2. **Check the 4 m** using its area.



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03 End point versus displacement

In a simplified tracking calculation, a probe starts at (**-100**, **0**) km and moves with constant velocity (**4**, **3**) km/s for **20 s**. The origin marks a stationary reference point. **Find final coordinates**, displacement magnitude and final distance from the origin. Plot the route.



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04 Add bounds

For the probe in question **3**, the initial x coordinate has a bound of ±**1 km** and each velocity component has an independent bound of ±**0.1 km/s**. Initial y and time are exact. **Find the final x** and y intervals.

Coordinate	Central (km)	Lower (km)	Upper (km)
x	____	____	____
y	____	____	____

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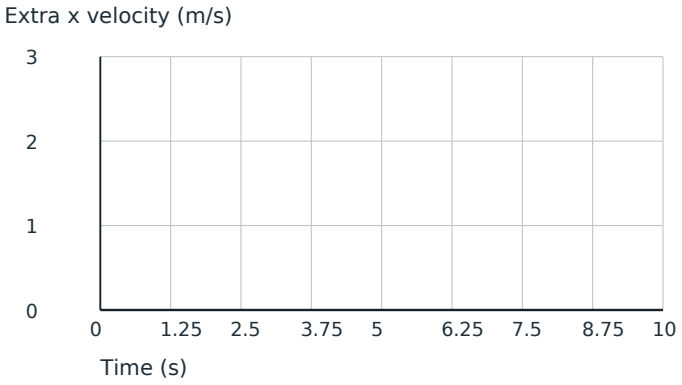
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Continue down each column.

05 A new force

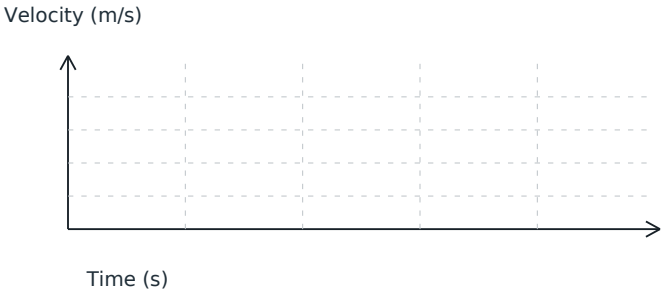
A **1000 kg** probe experiences a constant **200 N** net force in +x for **10 s**. Compared with coasting from the same initial state, find the extra x velocity and extra x displacement. **Use an acceleration**-time or velocity-change graph to check.



Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A car starts at **+35 m/s** and brakes at **-5 m/s²** until rest. A student extends that acceleration to **9 s**. **Find the stop time**/distance and **explain why extension** is invalid for ordinary braking.
Sketch the velocity–time graph; **label the given values**.



Marzano check - circle: ① ② ③

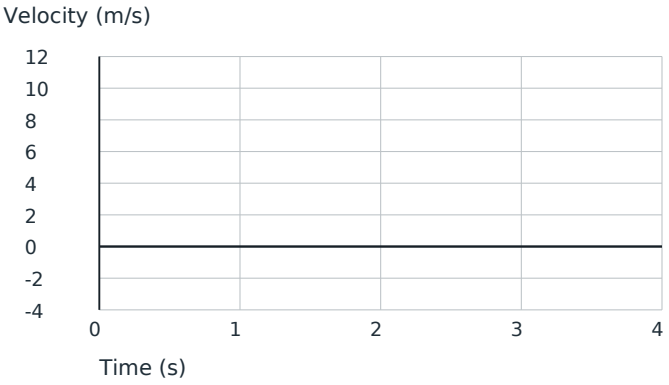
07 Exit check

Write a short explanation of questions **3** and **4** for the public. Include the **predicted position and coordinate ranges**. Explain **two reasons** these simplified calculations cannot establish a real asteroid collision forecast.

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A cart starts at x=**3 m** with v=**-2 m/s**. Its mass is **2 kg** and constant net force is **+6 N** for **4 s**.
1. **Predict final velocity** and position.
2. **Check whether** it turns.



Marzano check - circle: ① ② ③